EXP.NO: 3	Write a Servlet program that prints "Hello, World!" when accessed through a browser.

AIM:

To write a basic Servlet program that demonstrates the use of Java Servlets by printing "Hello, World!" when accessed through a browser.

ALGORITHM:

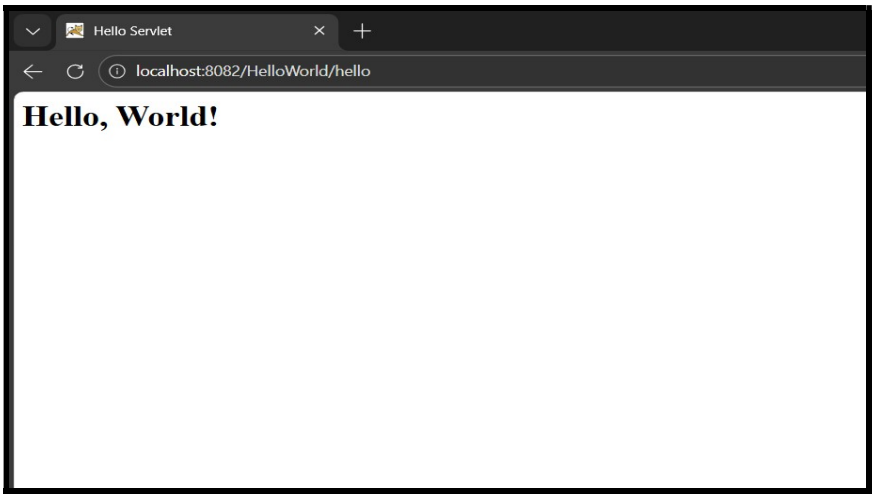
1. Input:

- The user makes an HTTP request (usually by entering a URL) through a browser.

2. Servlet Program Flow:

1. **Create a Servlet Class:**
   - Define a Java class that extends the HttpServlet class.
   - Override the doGet() method, which handles HTTP GET requests from the client (browser).
2. **Handle HTTP Request:**
   - In the doGet() method:
     - Get the **PrintWriter** object to write the response to the browser.
     - Set the response content type to **text/html**.
     - Write the **"Hello, World!"** message to the response.
3. **Compile the Servlet Class:**
   - Compile the Java Servlet file using the javac command to generate a .class file.
4. **Deploy the Servlet:**
   - Place the compiled servlet class file in the appropriate directory of the web application (typically in the /WEB-INF/classes folder).
   - Configure the servlet in the web.xml file of the web application or use annotations.
5. **Access the Servlet via Browser:**
   - Start the **Apache Tomcat** or another servlet container.
   - Access the servlet by entering the appropriate URL (e.g., http://localhost:8082/HelloWorld/hello).

Output:



Java File (HelloWorldServlet.java) :

```
import java.io.*;

import jakarta.servlet.*;

import jakarta.servlet.http.*;


public class HelloWorldServlet extends HttpServlet {

   public void doGet(HttpServletRequest request,HttpServletResponse response) throws
IOException,ServletException {

      response.setContentType("text/html");

      PrintWriter out= response.getWriter();

      out.println("<html>");

      out.println("<head><title>Hello Servlet</title></head>");

      out.println("<body>");

      out.println("<h1>Hello, World!</h1>");

      out.println("</body>");

      out.println("</html>");

   }

}
```

Java Compiled File (HelloWorldServlet.class)

```
// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij
IDEA).

import jakarta.servlet.ServletException;

import jakarta.servlet.http.HttpServlet;

import jakarta.servlet.http.HttpServletRequest;

import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;

import java.io.PrintWriter;
```

```
public class HelloWorldServlet extends HttpServlet {
  public HelloWorldServlet() {
  }

  public void doGet(HttpServletRequest var1, HttpServletResponse var2) throws
IOException, ServletException {
    var2.setContentType("text/html");
    PrintWriter var3 = var2.getWriter();
    var3.println("<html>");
    var3.println("<head><title>Hello Servlet</title></head>");
    var3.println("<body>");
    var3.println("<h1>Hello, World!</h1>");
    var3.println("</body>");
    var3.println("</html>");
  }
}
```

File (web.xml) :

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
        https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
        version="6.0">

   <servlet>
      <servlet-name>HelloWorld</servlet-name>
      <servlet-class>HelloWorldServlet</servlet-class>
   </servlet>
```

```
    <servlet-mapping>
        <servlet-name>HelloWorld</servlet-name>
        <url-pattern>/hello</url-pattern>
    </servlet-mapping>

</web-app>
```

RESULT:

The browser will display the message "Hello, World!" returned by the Servlet.